

Note that this Sample Answer is for the modified version of the Questions as prepared for Coursera. The ordering and some wording has changed – and a new section added.

Thank you for your patience as we developed this problem and answer.

Additionally, the answers in blue are from the model example Professor Egger developed for these questions below. Your model will be different and thus your exact answer will be different. Use the model answers ONLY as a guide to help you develop your own model.

Case Study
Modeling Credit Card Default Risk
and Customer Profitability
Draft as of 10.8.15

Part 1: Building Your Own Binary Classification Model

1.0 Introduction

You work for a bank as a business data analyst in the credit card risk-modeling department. Your bank recently conducted a bold experiment: over a short time interval three years ago, it quietly issued 600 credit cards to everyone who applied, regardless of their credit risk.ⁱ

After three years, 150, or 25%, of card recipients defaulted – they failed to pay back at least some of the money they owed. However, the bank collected very valuable proprietary data that it can now use to optimize its future card-issuing process.ⁱⁱ

The bank initially collected six pieces of data about each person.

Age,
Years at current employer,
Years at current address,
Income over the past year,
Current credit card debt, and
Current automobile debt.

You are first asked to propose a binary classification model for default that uses only data from one or more of the above six inputs, and outputs a single “score.” The relative rank-ordering of scores will determine the model’s effectiveness. For convenience, you are asked to use a scale for your score that has a maximum < 3.5 and a minimum > -3.5 .ⁱⁱⁱ

Initially you are not told what the bank’s best estimate for cost per False Negative (accepted applicant who becomes a defaulting customer) and False Positive

(rejected customer who would not have defaulted). Therefore, the best you can do is to design a model that maximizes the Area Under the ROC Curve, or AUC.

You are told that if your model is effective (“high enough” AUC – not defined) and “robust” (not defined, but in general means relatively little change in AUC across multiple sets of available data) that it may be adopted by the bank as a predictive model for default, to determine which future applicants will be issued credit cards.

1.1 First Binary Classification Model

You are first given a “training set” of 200 out of the 600 people in the experiment. Design your model on this set. Standardize your data first. You may combine the six inputs by adding them to or subtracting them from each other, taking simple ratios, etc – The only restriction is that your final “score” needs to be scaled so that the maximum is less than 3.5 and the minimum is greater than -3.5, so you can use the Excel “AUC Calculator” provided.

Q: What is your model? Give it as a function of the six inputs that outputs a single numerical score between -3.5 and 3.5 for each applicant. *[No exact answer]*

DE Model (as example):

(Model 1: $[-(\text{cc debt} + \text{auto debt})/\text{income}] * 6 - 2.9$)
+ Model 2: $[-(\text{standardized age} + \text{years at employer} + \text{years at address})/2] * .6$

Q: What is your model’s AUC on the Training Set? *[No exact answer]*

.76

(at threshold of -.75, with cost per test of \$863 – see below).

1.2 Initial Assessment for Over-fitting (testing your model on New Data)

Next test your model, without changing any parameters, on the Test Set of 200 additional applicants.

Q: What is your model’s new AUC on the *Test Set*? *[No exact answer]*

.78

(with a cost per test at -.75 of \$913 – see below)

If the test set AUC is much worse than the training set AUC, consider making your model simpler – probably using fewer input variables – as it is probably over-fitting the training set data.

Ideally, you would like a model to have only slightly lower performance on “new” data on which it was not trained. If you can develop a model exclusively on training set data that has an AUC over [.72] and the training set data also shows an AUC over [.68], you are doing very well. But don’t over-optimize your model on the test set data either, or it will turn out to be over-fitted on new, “validation set” data.

1.3 Finding the Cost-Minimizing Threshold for your Model

Now that you have, hopefully, developed your model to the point where it is relatively “robust” across the training set and test set, your boss at the bank finally gives you its current rough estimate of the bank’s average costs for each type of classification error.^{iv}

Cost Per False Negative: \$5000

Cost Per False Positive: \$2500

Note that for the 600 individuals that were automatically given cards without being classified, the total cost of the experiment turned out to be $25\%(\$5000)*600$ or \$7.5 million. This is \$1,250 per event. Only models with lower cost per event than this have *any* value.

Q: On the *training set*, what is the threshold score for your current classification model that minimizes costs per event on the training set? *[No exact answer]*

at threshold -.75,

Q: What is your minimum cost per event on the *training set*? *[No exact answer]*

\$863

Q: At that *same* threshold score (NOT the threshold score that would minimize costs for the new Test Set, but the “old” threshold score that minimized costs on the Training Set) what is the cost per event on *the test set*? *[No exact answer]*

\$913

If you find that your costs per event on the test set are much higher than your costs per event on the training set, consider making your model simpler – probably using fewer input variables – as it is probably still over-fitting the training set data. Problems that were not obvious at the ROC-curve stage may emerge when minimizing costs.

1.4 Putting a Dollar Value on Your Model Plus the Data

Again assume *Test Set* results are sustainable long term.

Q: How much money does the bank save, per event, using your model and its data-inputs, instead of issuing credit cards to everyone who asks? *[No exact answer]*

\$1250 – \$913 =

\$338/test

Q: Given that it apparently cost the bank \$750,000 to conduct the three-year experiment, how many applicants will need to be reviewed using your model before the bank pays back its investment?

750,000/338 =

2,222 tests

1.5 Confusion Matrix Metrics at the cost-Minimizing Threshold for your Model

On the *test set*,

Q: What are your model's False Positive and True Positive rates? *[No exact answer]*

.26, .66

Q: What is the model's Positive Predictive Value (PPV) and Negative Predictive Value (NPV)? *[No exact answer]*

.46, .87

Q: What is the “test incidence” of your test, on the test set, at the threshold from the training set? In other words, what percentage of applicants does your model classify Positive as “defaulters” and what percentage Negative as “non-defaulters”? *[No exact answer]*

33% Classified as Defaulters, 67% classified as non-defaulters

ⁱ Note that this is equivalent to setting a very high threshold for classification as “Positive” - so high that there are no True Positives (rejected applicants who would have defaulted) and no False Positives (rejected applicants who would not have defaulted), only True Negatives (non-defaulter customers) and False Negative (defaulter customers).

ii The 600 people are divided into three groups of 200 each, using a randomizing procedure that maintains the “base rate” of default at a constant 25% over all three groups. The first set of 200 is called the “Training Set.” It is for experimenting, and optimizing your model. The second set of 200 is called the “Test Set.” It is for evaluating how your model performs when it is faced with new data. It is essential that your initial model not take into account any properties of the Test set data, except the base rate of default. The third set of 200 is called the “Validation Set.” After any initial over-fitting has been eliminated, your model’s performance on new data from the validation set should be only slightly worse than its performance on the test set.

iii An example of a model on the training set would be $(\text{standardized age}) + (\text{standardized years at employer})/1.7$. Because (standardized age) ranges from a Max of +2.56 to a Min of -1.76, and (standardized years at employer) ranges from a Max of 3.4 to a Min of -1., division by 1.7 ensures that no sums could exceed the 3.5 Maximum. Note that it is not required that you weigh each input the same – although this is often sufficient to build a very good model. And you are free to take ratios of inputs – for example $(\text{years at current job})/(\text{years at current address})$ if you think those ratios will yield more information than the individual values. The main thing is simply to experiment and try a lot of different combinations, and see what works and keeps working on new data (the Test Set). This is the best way to develop intuition about what is actually going on in model-building.

iv Note that all bank models here include only profits and losses within three years of when a card is issued, so the impact of out-years (years beyond 3) can be ignored.